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Cracking PRMT5: Mechanistic Insights, Clinical Advances, and AI-Driven Strategies

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Abstract

Protein arginine methyltransferase 5 (PRMT5) is a type II methyltransferase that catalyzes symmetric arginine dimethylation on histone and non-histone proteins, thereby modulating transcription, RNA splicing, and diverse signaling pathways. Dysregulated PRMT5 activity contributes to tumorigenesis and multiple pathological conditions, positioning it as a high-priority therapeutic target with strong translational relevance. Current inhibitor strategies encompass S-adenosylmethionine (SAM)-competitive agents, substrate-competitive and substrate-dependent modulators, dual-binding and methylthioadenosine (MTA)-cooperative inhibitors, PROTAC-based degraders, repurposed FDA-approved drugs, and rational combination regimens designed to improve efficacy or overcome resistance. These approaches differ in mechanism of action, tumor specificity, and toxicity profiles. This review highlights their mechanistic underpinnings, clinical development, and inherent limitations, while also examining the potential application of PRMT5 inhibitors in non-oncologic indications, including ocular and cardiovascular diseases. Finally, it explores the role of biomedical informatics and artificial intelligence (AI) in drug repurposing and outlines future directions for advancing PRMT5-targeted therapeutics.

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Author TL is the founder of EQon Pharmaceuticals, LLC, the company that holds the licensed patent rights to the PRMT5 inhibitor PR5-LL-CM01 as well as the repurposed drugs candesartan and cloperastine for cancer treatment.

Keywords

arginine methylation; artificial intelligence; clinical trial; drug; PRMT5

1. Introduction

Protein arginine methylation is a critical post-translational modification (PTM) that regulates diverse cellular processes, including gene expression, RNA processing, and signal transduction [1]. This modification is mediated by a family of nine protein arginine methyltransferases (PRMTs), which are classified according to the type of methylation they catalyze [2]. Type I PRMTs (PRMT1, 3, 4, 6, and 8) generate asymmetric dimethylarginine, while PRMT7, the sole Type III enzyme, catalyzes monomethylation. By contrast, PRMT5, the prototypical Type II PRMT, has emerged as a key focus of cancer research owing to its elevated expression in numerous malignancies and its pivotal role in disease progression [1, 3].

PRMT5 catalyzes symmetric dimethylation of arginine residues on histone and non-histone proteins, thereby influencing chromatin remodeling, transcriptional repression, RNA splicing, and intracellular signaling [4]. Histone targets such as H4R3 and H3R8 mediate epigenetic gene silencing, while non-histone substrates regulate mRNA processing, cell cycle progression, and other essential functions [5]. Through these diverse activities, PRMT5 functions as a central regulator of tumor cell biology [6].

Aberrant overexpression or hyperactivation of PRMT5 has been documented across a wide range of cancers, including glioblastoma, lung, pancreatic, colorectal, prostate, and ovarian malignancies [7–10]. Oncogenic roles of PRMT5 are linked to methylation of nuclear factor (NF)-κB, repression of tumor suppressor genes, altered splicing of regulatory transcripts, and facilitation of tumor immune evasion [11]. Clinically, high PRMT5 expression correlates with poor prognosis in multiple tumor types [12]. Given its central role in tumorigenesis, PRMT5 has become an attractive therapeutic target [13]. Multiple classes of PRMT5 inhibitors have been developed, each characterized by distinct mechanisms of action and therapeutic potential [6]. This review provides an overview of the current landscape of PRMT5-targeted therapies, emphasizing their molecular mechanisms, clinical development, and potential applications beyond oncology. It also highlights the emerging role of biomedical informatics and artificial intelligence in drug repurposing, as well as the challenges and opportunities in translating PRMT5 inhibition into effective therapeutic strategies.

2. PRMT5 Inhibitor Categories and Mechanisms

PRMT5 inhibitors fall into seven categories (Fig. 1; Table 1), which we outline in more detail below.

2.1. SAM-Competitive Inhibitors:

S-adenosylmethionine (SAM) is the universal methyl donor utilized by all protein methyltransferases, including PRMT5. SAM-competitive inhibitors are small molecules designed to bind directly to the SAM-binding site of PRMT5, thereby preventing the methylation of both histone and non-histone substrates [14]. These inhibitors occupy the active site of the enzyme and disrupt the formation of methylated arginine residues that are essential for the regulation of various oncogenic processes [13].

PRMT5's catalytic domain contains two key binding pockets: one for SAM and another for the arginine residue of its substrate peptide. This structural feature provides two main avenues for drug development, targeting either the SAM-binding site or the substrate-binding pocket [14].

One of the early compounds, EPZ004777, a known DOT1L (disruptor of telomeric silencing 1-like) inhibitor, demonstrated weak PRMT5 inhibition and served as a starting point for nucleoside-based PRMT5 inhibitors [14]. Structure-activity relationship (SAR) efforts led to compound DS-437, which showed dual inhibition of PRMT5 and PRMT7 but lacked selectivity [13]. Further optimization yielded LLY-283 (Table 1), which achieved nanomolar potency (IC_{50} = 9.5 nM) against PRMT5 and exhibited favorable pharmacokinetics. Its binding to the SAM pocket was confirmed via co-crystal structure [14]. Halogen substitution at the phenyl moiety resulted in PF-06939999 (Table 1) [13]. JNJ64619178 (Table 1) developed by Janssen, was designed as a bisubstrate inhibitor that extends from the SAM-binding site into the substrate pocket, enhancing both potency and selectivity [14, 15]. Other examples like PRT543, PRT811 also belong to this category (Table 1, Fig. 2A).

SAM-competitive inhibitors offer a direct and specific approach to inhibit PRMT5 enzymatic activity and have shown favorable selectivity profiles in some cases. However, their development faces key challenges. The SAM-binding pocket is highly conserved among methyltransferases, complicating selectivity [14]. Additionally, the high intracellular concentration of SAM can outcompete inhibitors, potentially reducing their effectiveness [14]. Furthermore, the clinical tolerability of some early inhibitors has been limited by toxicities in normal proliferating tissues [13]. These challenges highlight the importance of careful optimization in structure and pharmacokinetics to achieve effective and selective PRMT5 inhibition in vivo [13, 14].

2.2. Substrate-Competitive Inhibitors:

Substrate-Competitive inhibitors (Fig. 1, Fig. 2B, Table 1) represent a distinct class of PRMT5-targeting compounds that rely on the presence of SAM for their activity. Unlike SAM-competitive inhibitors that directly compete with the methyl donor, these molecules bind to a secondary pocket created only after SAM is bound, forming a PRMT5–SAM–inhibitor ternary complex that stabilizes the enzyme in an inactive conformation [16]. One of the earliest and most studied examples is EPZ015666 (GSK3235025), which demonstrated strong biochemical and cellular potency in lymphoma models [17]. Subsequent optimization led to EPZ015938, also known as GSK3326595, which advanced into Phase I clinical trials for solid tumors and non-Hodgkin lymphoma [18] (Table 2).

Structural analyses have shown that substrate-competitive inhibitors engage the substrate-binding groove rather than the SAM pocket. Binding involves key π–π stacking interactions with conserved residues such as Phe327, essential for PRMT5's symmetric dimethylation activity [14]. These inhibitors mimic the substrate arginine side chain, blocking methylation without competing with SAM itself [19].

Substrate-competitive inhibitors offer several advantages. Their dependence on the PRMT5–SAM complex enhances selectivity, reducing off-target effects on other PRMTs [15]. They are particularly effective in MTAP-deleted cancers, where elevated methylthioadenosine (MTA) levels sensitize cells to PRMT5 inhibition. Moreover, they maintain activity even in high-SAM cellular environments, overcoming a major limitation of SAM-competitive agents [2]. However, these inhibitors also present challenges. Their activity is contingent upon SAM binding, which may vary by cellular context. Designing such compounds requires precise structural alignment due to the induced-fit nature of their binding pocket [2]. Additionally, although clinical-stage molecules like GSK3326595 have demonstrated improved pharmacokinetics, achieving optimal bioavailability and tissue distribution remains a hurdle for broader application [18].

In summary, SAM-uncompetitive inhibitors provide a powerful and selective strategy to inhibit PRMT5 activity. Continued optimization, including formulation enhancements such as nanocrystals, may expand their therapeutic potential across a wider range of PRMT5-driven malignancies [20].

2.3. Substrate-Dependent Inhibitors:

Our group has developed PR5-LL-CM01, a novel PRMT5 inhibitor that employs a distinctive substrate-dependent binding mechanism (Fig. 1, Fig. 2C, Table 1) [8]. Unlike classical inhibitors that interact exclusively with a single binding pocket, the activity of PR5-LL-CM01 is influenced by the presence or absence of the cofactor SAM. Specifically, PR5-LL-CM01 engages the SAM-binding pocket when SAM is absent, but in the presence of SAM it adopts an alternative binding mode at a secondary site within PRMT5. This context-dependent interaction provides unique pharmacological advantages. By shifting its binding site according to SAM availability, PR5-LL-CM01 may achieve greater target selectivity, thereby reducing off-target activity against other methyltransferases that share similar SAM-binding motifs. In addition, this flexible binding mode could help overcome resistance mechanisms that often arise with inhibitors restricted to the SAM-binding site, broadening its therapeutic potential. At the same time, the properties that distinguish PR5-LL-CM01 also introduce important challenges. The complex binding dynamics associated with substrate dependence make it more difficult to predict or model inhibitor activity, since efficacy may fluctuate with intracellular SAM concentrations and the conformational state of PRMT5. Moreover, tissue-specific variations in SAM availability could influence potency and contribute to differential toxicity profiles, requiring thorough pharmacological characterization in preclinical studies. Despite these complexities, PR5-LL-CM01 represents a promising advance in the PRMT5 inhibitor landscape, as its novel mechanism offers a rational strategy for improving selectivity and achieving more durable responses.

Notably, the principle of context-dependent inhibition has already proven successful in oncology. Several FDA-approved kinase inhibitors, including imatinib (BCR-ABL), dasatinib (BCR-ABL/Src), nilotinib (BCR-ABL), and bosutinib (Src/BCR-ABL), demonstrate flexible binding modes that vary with ATP occupancy. These precedents highlight how substrate-dependent mechanisms can be successfully translated into the clinic, providing a strong rationale for further development of PR5-LL-CM01 as a next-generation PRMT5 inhibitor for cancer therapy.

2.4. Dual-site (SAM + substrate) inhibitors:

One example is HLCL65 (Fig. 1, Fig. 2D, Table 1) which embodies a dual-site pharmacophore designed to engage interactions within both the cofactor (SAM) pocket and the substrate-recognition groove of PRMT5 simultaneously [21]. By spanning these adjacent sites, HLCL65 sterically and electronically blocks methyl transfer: it occludes the SAM donor position while also preventing productive binding/positioning of the arginine substrate. This bivalent engagement increases apparent affinity through avidity and stabilizes an inactive PRMT5 conformation, leading to potent suppression of symmetric arginine dimethylation on histone and non-histone targets.

The pros of the dual-site design include (i) high potency and durability of inhibition via cooperative binding across two pockets; (ii) enhanced selectivity for PRMT5 over other PRMTs by exploiting a composite interface; and (iii) a higher barrier to resistance, since simultaneous mutations at both sites are less likely to arise without fitness cost. Key challenges are medicinal-chemistry and ADME related: dual-site scaffolds can be larger and more polar, risking reduced cell permeability, oral exposure, and tissue penetration, and they may face efflux or metabolic liabilities; additionally, any SAM-pocket contacts must avoid cross-reactivity with other methyltransferases. Future development of HLCL65 will likely benefit from structure-guided optimization aimed at improving its pharmacological properties. Strategies such as linker refinement, conformational restriction or macrocyclization, and prodrug design could be applied to enhance cell permeability, metabolic stability, and overall pharmacokinetics (PK)/pharmacodynamics (PD) performance, thereby broadening the therapeutic index. At the same time, the integration of robust pharmacodynamic biomarkers, including global SDMA levels and histone H4R3me2s, will be essential for accurately monitoring target engagement and guiding dose selection in both preclinical and clinical studies. Together, these approaches will not only sharpen HLCL65's drug-like profile but also provide a stronger translational framework for its progression into advanced therapeutic development.

2.5. MTA-Cooperative Inhibitors:

Methylthioadenosine phosphorylase (MTAP) is an essential enzyme in the methionine salvage pathway, catalyzing the breakdown of 5′-methylthioadenosine (MTA) into adenine and methionine [22, 23]. In many human cancers, MTAP is co-deleted with the tumor suppressor gene Cyclin-Dependent Kinase Inhibitor 2A (CDKN2A), resulting in loss of MTAP function and leading to the accumulation of MTA in tumor cells [22, 24] (Fig. 3). Approximately 15% of all solid tumors, including non-small cell lung cancer, gliomas, and pancreatic cancers, exhibit MTAP deletion (MTAP-del) [25]. Importantly, MTA is a structural analog of SAM and functions as a competitive inhibitor of PRMT5, creating a partial inhibition, or hypomorphic state, in MTAP-deleted cells [20].

This biological vulnerability has been exploited for the development of MTA-cooperative PRMT5 inhibitors (Fig.1, Fig. 2E, Fig. 3, Table 1, Table 2). Examples include AM-9747, AMG193, AZD-3470, BAY 3713372, BGB-58067, BMS-986504 (MRTX1719), CTS3497, IDE397, PEP08, S095035, TNG456, TNG462, TNG908. These compounds are designed to selectively bind and inhibit PRMT5 in the presence of elevated MTA levels, achieving preferential targeting of MTAP-deleted cancer cells while sparing normal, MTAP-wildtype (MTAP-WT) tissues [26] (Fig. 3). Unlike first-generation PRMT5 inhibitors, which show systemic toxicity due to non-selective inhibition, MTA-cooperative inhibitors are expected to offer an improved therapeutic window [27]. One such example is AM-9747, a quinolin-2-amine-based inhibitor identified through DNA-encoded library screening, AM-9747 binds cooperatively with MTA at the PRMT5 active site [27]. Structural studies revealed interactions with PRMT5 glutamate-444 and hydrophobic contacts with MTA, leading to potent inhibition of symmetric dimethylarginine (SDMA) formation. AM-9747 demonstrated a 75-fold selectivity favoring MTAP-del over MTAP-WT cells and achieved a >80% reduction in SDMA levels in xenograft models [27]. Another example is AMG193. Developed by Amgen, AMG193 is the first MTA-cooperative PRMT5 inhibitor to enter clinical evaluation. It exhibits strong inhibition of PRMT5 activity selectively in MTAP-deleted cells while sparing normal tissues. Preclinical models showed robust tumor growth inhibition without affecting MTAP-WT xenografts [24].

By leveraging the unique metabolic state of MTAP-deleted tumors, MTA cooperative PRMT5 inhibitors offer the potential for a broader therapeutic index and improved patient outcomes [28, 29]. Critically, preclinical data indicate that compounds such as BMS-986504 (MRTX1719), which maintain PRMT5 inhibition while sparing immune cells, could overcome the immunosuppressive drawbacks of global PRMT5 blockade [24]. However, challenges remain. The full impact of these inhibitors on tumor-infiltrating immune cells requires further examination, especially given PRMT5's roles in immune modulation [30].

2.6. PROTAC inhibitors:

Proteolysis-targeting chimeras (PROTACs) targeting PRMT5 represent a novel therapeutic approach that extends beyond conventional enzymatic inhibition. Unlike classical PRMT5 inhibitors that block the SAM-binding site, PRMT5 PROTACs harness the ubiquitin–proteasome system to induce selective degradation of the protein, thereby abolishing both its catalytic activity and non-enzymatic scaffolding functions.

Several groups have reported first-in-class PRMT5 degraders (Fig. 1, Fig. 2F, Table 1): Shen et al. described MS4322, a VHL-recruiting PROTAC that degrades PRMT5 in cancer cell lines and showed proteasome- and E3-dependent activity in vitro and in vivo proof-of-concept studies [31]. A 2024 study identified YZ-836P, a cereblon (CRBN)-recruiting PRMT5 degrader with cytotoxic activity in triple-negative breast cancer models, demonstrating that different E3 ligases (VHL vs CRBN) can be used to design PRMT5 degraders with distinct cellular profiles [32]. Early proof-of-concept studies have demonstrated efficient PRMT5 degradation, leading to stronger suppression of downstream methylation and enhanced antiproliferative effects compared to standard inhibitors.

The major advantages of PRMT5 PROTACs include sustained target knockdown, the ability to overcome resistance linked to mutations in the SAM-binding pocket, and potentially improved selectivity due to catalytic turnover of the degrader. However, challenges remain, including optimization of PK and oral bioavailability, the risk of off-target degradation through non-specific E3 ligase recruitment, and the need to fully assess toxicity given PRMT5's essential role in normal RNA splicing and development. Overall, PRMT5 PROTACs are at an early but promising stage, offering a powerful alternative to traditional inhibition strategies, though significant preclinical validation and safety profiling are still required.

2.7. Repurposed PRMT5 Inhibitors:

Cloperastine hydrochloride, candesartan cilexetil, and tadalafil are FDA- or EMA-approved drugs identified as potential PRMT5 inhibitors (Fig. 1, Fig. 2G, Table 1). Cloperastine (cough suppressant) and candesartan (antihypertensive) were identified through AlphaLISA screening and have been shown to reduce NF-κB methylation and activation, inhibit cancer cell proliferation, and suppress tumor growth in xenograft models, all with minimal toxicity [9]. Tadalafil (Phosphodiesterase type 5, PDE5 inhibitor) was identified through virtual screening and confirmed by SPR binding, inhibiting PRMT5-mediated methylation of histone H4 and AlkB Homolog 7 (ALKBH7) in cells. These repurposed agents highlight a promising and accelerated approach for targeting PRMT5 in anticancer therapy [33].

Repurposed PRMT5 inhibitor candidates offer several advantages, most notably their pre-established safety profiles, which include well-characterized PK profile, toxicity, and dosing parameters, potentially accelerating clinical translation. This approach is also cost-effective and time-saving compared to the development of entirely new inhibitors, and some candidates, such as cloperastine and candesartan, have demonstrated functional inhibition of PRMT5 activity in cellular, and in certain cases, animal models [9]. However, these repurposed agents also have notable limitations. Their potency is generally modest, with IC_{50} values and effective concentrations often in the micromolar range, making them less potent than rationally designed PRMT5 inhibitors. Additionally, clinical data for oncological indications are lacking, as no trials have yet evaluated these drugs for cancer specifically via PRMT5 inhibition. Finally, potential off-target effects should be considered, given these drugs' primary activities, such as cardiovascular effects for candesartan or PDE5 inhibition for tadalafil, which may require careful monitoring.

Nevertheless, given the many examples of drugs successfully repurposed for oncology, such as thalidomide for multiple myeloma, metformin for breast and pancreatic cancers, aspirin for colorectal cancer prevention, and itraconazole for Hedgehog-driven tumors, these agents hold strong potential for successful development in cancer therapy.

3. Combination Strategies and the Clinical Development Landscape

PRMT5 inhibitors have shown promise, particularly in MTAP-deleted tumors, but single-agent efficacy is often limited. Combining these inhibitors with chemotherapy, targeted therapies, or immunotherapies may boost effectiveness and overcome resistance. Here, we summarize current combination strategies and their progress in clinical development.

3.1. Combination strategy:

PRMT5 inhibitors are increasingly being explored in combination therapies to enhance anticancer efficacy, overcome resistance, and exploit synthetic lethality. One of the most widely studied approaches is pairing PRMT5 inhibitors with DNA-damaging chemotherapies such as gemcitabine or cisplatin [34]. Because PRMT5 regulates RNA splicing and DNA damage response pathways, its inhibition sensitizes tumor cells to genotoxic stress, showing promising results in pancreatic, lung, and ovarian cancer models [1, 35]. Another strategy combines PRMT5 inhibition with PARP inhibitors, which exploits homologous recombination deficiencies in BRCA-mutant tumors, inducing synthetic lethality in breast and ovarian cancer models [36]. Additionally, Brown-Burke et al. provide a mechanistic rationale for combining PRMT5 inhibition with venetoclax, a BCL-2 inhibitor, in mantle cell lymphoma (MCL). PRMT5 inhibition disrupts splicing and expression of key apoptotic regulators, sensitizing tumor cells to BCL-2 blockade, thereby enhancing apoptosis to single-agent therapy [37].

Combinations with immunotherapy are also being explored. By modulating the splicing of immune-related transcripts and reprogramming the tumor microenvironment, PRMT5 influences tumor immune evasion, making its inhibition a potentially synergistic strategy when paired with checkpoint inhibitors such as anti-PD-1 or anti-PD-L1 antibodies [38]. Additional strategies involve combining PRMT5 inhibitors with epigenetic modulators, such as EZH2 or DOT1L inhibitors, to disrupt transcriptional and chromatin regulation more effectively [39, 40]. Furthermore, PRMT5 has been shown to regulate cMYC stability and promote epithelial-mesenchymal transition (EMT) via pathways including EGFR/AKT/β-catenin signaling, highlighting its central role in tumor progression. Current research is actively exploring combination approaches that pair PRMT5 inhibitors with chemotherapeutic agents, as well as dual-targeting strategies involving PRMT5 and other oncogenic regulators such as PRMT1 [41]. In addition, repurposed FDA-approved drugs with PRMT5 inhibitory activity, such as cloperastine, candesartan, and tadalafil, offer a cost-effective means to enhance the efficacy of standard therapies in preclinical models [9]. Taken together, these strategies underscore the versatile therapeutic potential of PRMT5 inhibition and highlight multiple avenues for translational and clinical development.

3.2. Clinical Development Landscape:

Several PRMT5 inhibitors are currently being evaluated in clinical trials for a variety of solid tumors and hematologic malignancies, highlighting growing interest in targeting this epigenetic regulator. Table 2 provides a comprehensive summary of the current status of clinical trials evaluating PRMT5 inhibitors [42].

Early-phase studies focus on assessing safety, tolerability, PK, PD, and preliminary antitumor activity. For example, AZD3470 is being tested in patients with advanced or metastatic solid tumors to establish the maximum tolerated dose (MTD) and recommended Phase 2 dose. AMG 193, a cofactor-dependent PRMT5 inhibitor, is being evaluated in tumors with MTAP deletions in both monotherapy and combination therapy settings. Other agents, including IDE397, JNJ-64619178, and TNG908, are also under early-phase investigation in advanced solid tumors and select hematologic malignancies, with a particular focus on MTAP-deleted cancers.

Beyond monotherapy, several trials are exploring combination strategies and window-of-opportunity designs to enhance therapeutic efficacy. For instance, GSK3326595 is being studied in early-stage breast cancer to assess PD effects in tumor tissue. Across these studies, objectives include optimizing dosing regimens, identifying biomarkers of response, and establishing preliminary clinical activity. Collectively, these trials underscore the translational potential of PRMT5 inhibition and lay the groundwork for future combinations with chemotherapies, immunotherapies, or other epigenetic modulators to maximize clinical benefit across diverse cancer types.

Additionally, BMS-986504 is a selective PRMT5 inhibitor currently in a Phase I trial for patients with advanced or metastatic solid tumors and hematologic malignancies. The study aims to evaluate the safety, tolerability, MTD, PK/PD profiles, and preliminary antitumor activity of BMS-986504 as monotherapy. This trial contributes to the broader effort to explore PRMT5-targeted therapies in cancers with potential vulnerabilities, including MTAP-deleted tumors [14, 26, 28, 43].

Current clinical trials are heavily focused on MTA-cooperative inhibitors (Table 1, Table 2). Early results show promising activity in MTAP-deleted tumors, while other classes of PRMT5 inhibitors remain largely untested. This underscores the need to optimize safety, refine patient selection, and develop combination strategies to both diversify the types and structures of PRMT5 inhibitors evaluated and expand their applicability beyond MTAP-deleted cancers, which account for only approximately 15% of patients.

4. Expanding Therapeutic Horizons: Beyond Oncology

Beyond its extensive application in oncology, PRMT5 has shown promising potential in other disease contexts. Below, we highlight several examples of such applications (Fig. 4):

4.1. PRMT5 Inhibition in Inflammatory and Autoimmune Diseases:

PRMT5 is a key regulator of immune cell proliferation and function, making it a potential target for autoimmune disorders. In multiple sclerosis (MS), PRMT5 is upregulated in pathogenic Th1 cells, promoting disease progression. Pharmacological inhibition of PRMT5 suppresses Th1 proliferation and IL-2 secretion. Selective inhibitors such as HLCL65 have demonstrated efficacy in reducing disease severity in experimental autoimmune encephalomyelitis (EAE) models of MS [44].

PRMT5 inhibition has also been explored in inflammatory bowel disease (IBD). Elevated PRMT5 expression is observed in colonic tissues during active IBD and enteric infections. PRMT5 haploinsufficiency impairs the mucosal barrier and delays pathogen clearance in mice, indicating that carefully tuned PRMT5 inhibition could support mucosal healing without exacerbating infections [34]. However, most evidence to date is derived from experimental or animal models, and the complexity of immune regulation underscores the need for careful evaluation of safety, selectivity, and long-term effects before clinical translation can be established.

4.2. PRMT5 in Fibrotic Diseases:

PRMT5 contributes to fibrosis, a key factor in heart failure and other chronic conditions. In cardiac fibrosis models, PRMT5 promotes fibroblast activation and differentiation into myofibroblasts via Smad3 methylation, stabilizing its phosphorylation and amplifying TGF-β signaling. In vivo silencing or pharmacological inhibition of PRMT5 reduces fibrosis and improves cardiac function, suggesting a potential therapeutic role for PRMT5 inhibition in fibrotic diseases [44]. While these findings are promising, they remain largely preclinical, and further studies are needed to clarify safety, efficacy, and translatability before clinical application can be considered.

4.3. PRMT5 and Ferroptosis Regulation:

Ferroptosis, an iron-dependent form of cell death, is a promising cancer target. Tumor cells upregulate methionine metabolism to stabilize Glutathione Peroxidase 4 (GPX4) and resist ferroptosis. PRMT5 methylates GPX4 at arginine 152, preventing its degradation and counteracting ferroptosis. Combining PRMT5 inhibitors with ferroptosis inducers effectively restricts tumor growth, suggesting a novel combinatorial strategy for resistant cancers [4].

4.4. PRMT5 in Viral Infections:

PRMT5 facilitates viral replication by methylating viral proteins. For example, it methylates the VP1 polymerase of infectious bursal disease virus (IBDV) at arginine 426, enhancing polymerase activity and viral propagation. PRMT5 inhibition or knockout impairs VP1 function and reduces viral replication, offering a potential antiviral strategy [45].

4.5. PRMT5 in Ocular Diseases:

PR5-LL-CM01, a selective PRMT5 inhibitor developed by our group, suppresses NF-κB activation by reducing p65 dimethylation, thereby inhibiting inflammation and angiogenesis. In retinal endothelial cells, it blocks proliferation, migration, and tube formation without inducing apoptosis. In vivo, PR5-LL-CM01 reduces neovascular lesion size in a mouse model of neovascular age-related macular degeneration (nAMD), suggesting its potential role as a therapeutic target for ocular neovascular diseases, based on preclinical evidence [46].

Collectively, these studies illustrate that PRMT5 is a versatile therapeutic target beyond oncology, with potential applications in autoimmune, inflammatory, fibrotic, viral, and ocular diseases. While most evidence remains preclinical, strategic development of selective inhibitors and combination approaches may eventually inform therapeutic strategies for these conditions.

Biomedical Informatics and Artificial Intelligence in Drug Repurposing: To clinically target PRMT5 and associated biological activities such as the NF-κB pathway for cancer treatments, it is crucial to address the multifaceted heterogeneity of cancer patients, which arises from complex interactions among clinical characteristics, treatments, social determinants of health, tumor microenvironment dynamics, genetic and epigenetic alterations, and higher-order chromosome architecture. For example, spatial variability in PRMT5-related drug response among cancer cells within tumors fuels therapeutic resistance, disease relapse, and mortality. The rich and fast-growing national, regional, and local resources of clinico-spatio-omics (CliSpOmics) data across molecular, cellular, tissue, and population levels, as well as the advances of graph artificial intelligence (AI), provide a promising opportunity to address this need through comprehensive characterization and precision targeting of PRMT5 in cancer and beyond.

Noticeable CliSpOmics data resources that are intensively used for cancer drug repurposing include the Oncology Research Information Exchange Network (ORIEN) Avatar [47], the American Association for Cancer Research (AACR) Project GENIE® (Genomics, Evidence, Neoplasia, Information, Exchange) [48, 49], the All of Us project, the Human Tumor Atlas Network (HTAN) of the National Cancer Institute (NCI) [50, 51], the Genome Data Commons (GDC) of NCI [52], the National Clinical Cohort Collaborative (N3C, previously known as the National COVID Cohort Collaborative) [53], and others. Data and metadata standards enable the integration and harmonization of the CliSpOmics data from these public resources as well as local data repositories for drug repurposing. The Global Alliance for Genomics and Health (GA4GH) [54], for example, sets standards and frames policies for clinicogenomics data. These standards span from the data formatting such as the Pedigree specification, patient phenotyping standards and ontologies such as the Human Phenotype Ontology, the NCI Thesaurus, the Logical Observation Identifiers Names and Codes (LOINC) for clinical laboratory results, the Mondo Disease Ontology (Mondo), the genomics variation standards from the Variation Representation Specification (VRS) and Variant Annotation (VA) framework, and the NCI GDC data standard. Such fast-growing resources support biomedical informatics research for exploring the role of PRMT5 in cancers and other diseases and repurposing drugs targeting PRMT5-associated biological functions.

AI and machine learning approaches, particularly graph AI modeling, have been widely utilized to integrate multimodal and multiomics data for precision oncology, leveraging real-world data, clinical, demographic, socioeconomic, spatial, omics, single-cell, and imaging data. For example, the DeePaN [55] uses a graph convolutional network to integrate the electronic medical records (EMR), cancer registry, and tumor deep DNA sequencing data to identify patient subpopulations who are responsive to immune therapies. NK2 homeobox 1 (NKX2–1, as known as thyroid transcription factor 1 or TTF1) and KRAS mutations were hallmark signatures of the identified non-small cell lung cancer subgroup that responds to immune checkpoint inhibitors. NKX2–1 and PRMT4 expressions are negatively associated in lung adenocarcinoma [56]. SpaRx [57] (Fig. 5) utilizes adversarial graph AI to transfer the cell drug responses knowledge to spatial transcriptomics data to predict spatially differential drug responses to cancer treatments. SiGra [58], xSiGra [59], and spaCI [60] use explainable graph transformers to review spatial heterogeneity in biological functions and cell-cell communications among tumor tissues. SpaIM [61] uses graph style transfer to infer tumor spatial cellular activities. The OmiCLIP visual-omics foundation model integrates histopathologic imaging, spatial transcriptomics data, as well as clinicogenomics data to infer biological information. PRMT5 and associated biological activities, especially those in immune responses, demonstrate spatial heterogeneity of the distribution of PRMT5 expression, the activities of corresponding signaling pathways and biological functions, and cell-cell communications between immune cells and other cells. For example, PRMT5 and other protein arginine methyltransferases were reported to be highly expressed in tumor microenvironment of oral squamous cell carcinoma after the treatment with immunotherapy (nivolumab, a programmed death-1 (PD-1) inhibitor) [62]. Such spatial graph AI approaches are shifting the landscape of drug discovery, screening, and repurposing, bridging the gap between bench-side research and bedside clinical application, and accelerating the translational research of PRMT5.

5. Perspectives and Future Directions

Despite the promise of PRMT5 inhibitors, significant challenges remain before clinical translation. Systemic inhibition carries a risk of toxicity, particularly in proliferative tissues, highlighting the need for selective strategies such as exploiting tumor-specific vulnerabilities (e.g., MTAP deletion) or tissue-restricted delivery.

AI-driven tools, including SiGra, xSiGra, spaCI, SpaIM, and OmiCLIP, offer the potential to map cellular interactions, identify key niches, and guide context-specific inhibitor selection. However, realizing this potential requires rigorous validation of AI models, harmonization of diverse datasets, and responsible data integration. Together, these cutting-edge tools and AI-driven strategies have the potential to accelerate PRMT5 drug discovery and enable more precise, patient-tailored approaches. While clinical translation remains preliminary, these advances offer a promising path toward developing safe and effective therapies.

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Highlight:

- PRMT5 is a central regulator of gene expression and disease progression.
- Distinct classes of PRMT5 inhibitors show unique mechanisms and therapeutic promise.
- Offers the most up-to-date summary of current clinical trials.
- AI-driven strategies expand the potential for precise oncology therapies and applications beyond cancer.

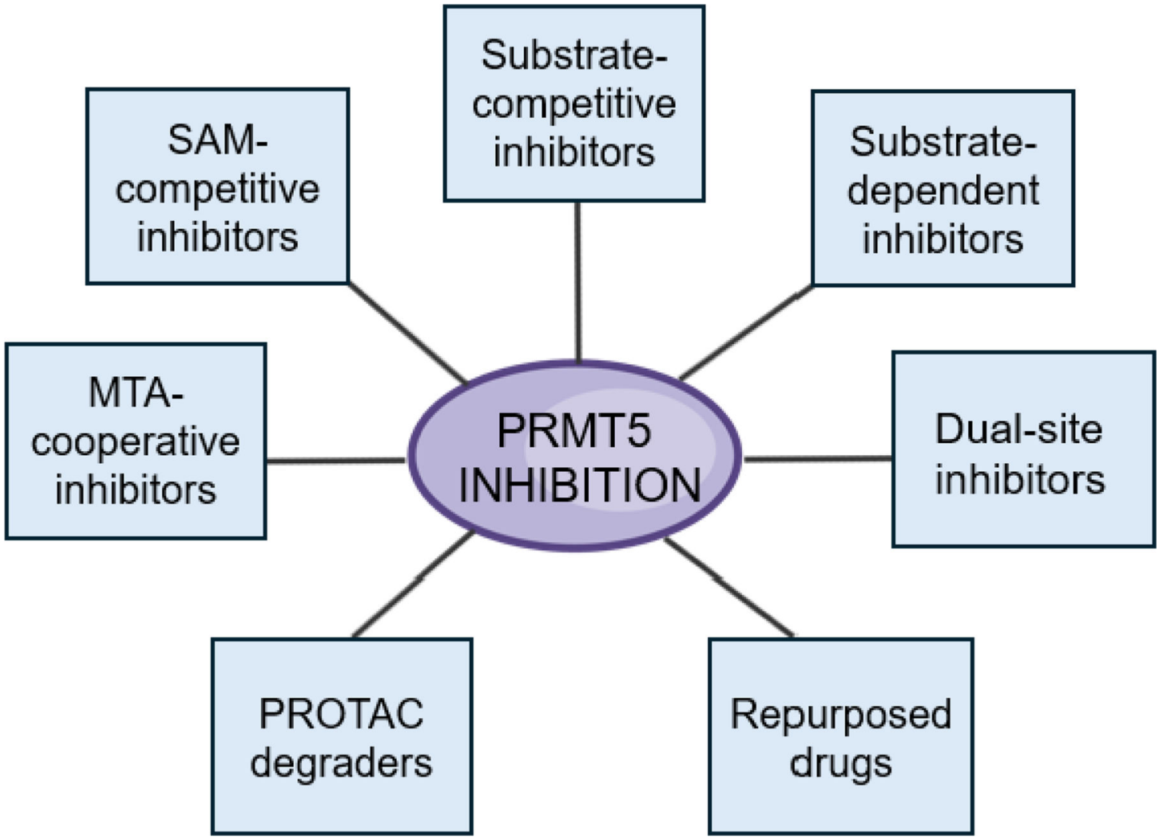

Fig. 1. Overview of PRMT5 Inhibition Strategies.

PRMT5 inhibitors can be categorized into 7 major classes based on their mechanism of action: SAM-competitive, substrate-competitive, substrate-dependent, dual-site, and MTA-cooperative inhibitors, PROTAC degrader, and repurposed drugs.

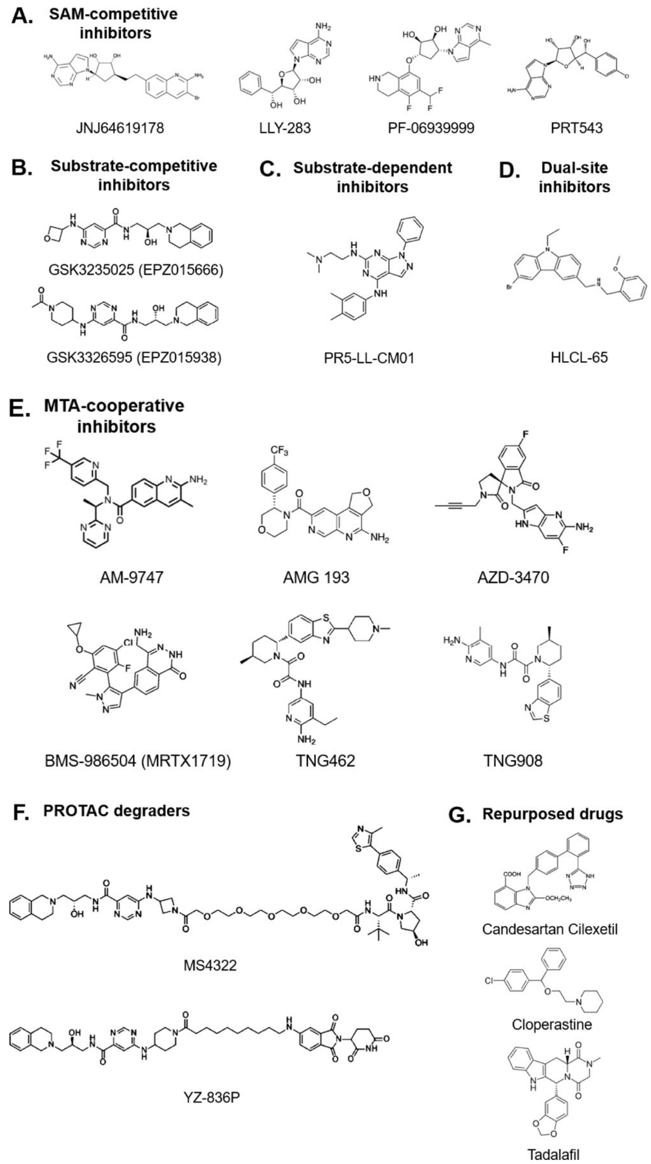

Fig. 2.

Chemical structures of some typical PRMT5 inhibitors.

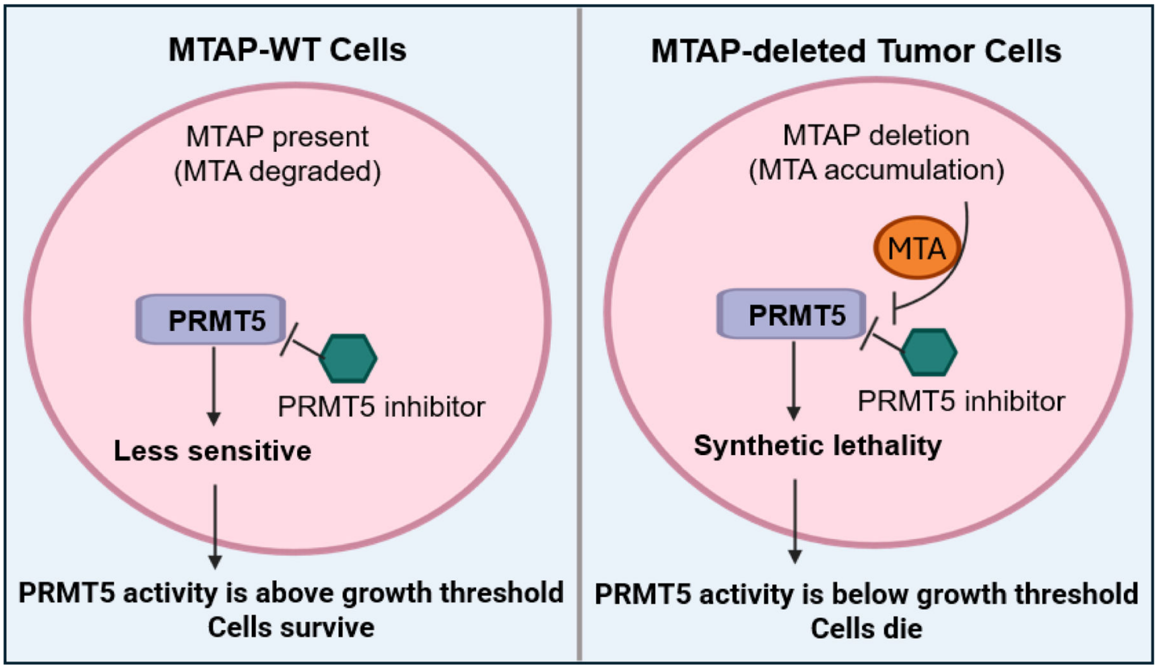

Fig. 3. Mechanism of MTA-cooperative PRMT5 inhibition in MTAP-deleted tumors.

In MTAP–wild-type (WT) cells, PRMT5 activity remains largely unaffected by PRMT5 inhibitors (left panel). In contrast, loss of MTAP in tumor cells leads to accumulation of methylthioadenosine (MTA) (right panel), which partially suppresses PRMT5 activity and creates a dependency on residual PRMT5 function. MTA-cooperative PRMT5 inhibitors (e.g., AMG193, MRTX1719, TNG908) exploit this metabolic vulnerability by selectively binding to PRMT5 in the presence of elevated MTA, thereby enhancing PRMT5 inhibition. This synthetic lethal interaction suppresses tumor cell proliferation while sparing normal MTAP-WT tissues. Created with BioRender.com.

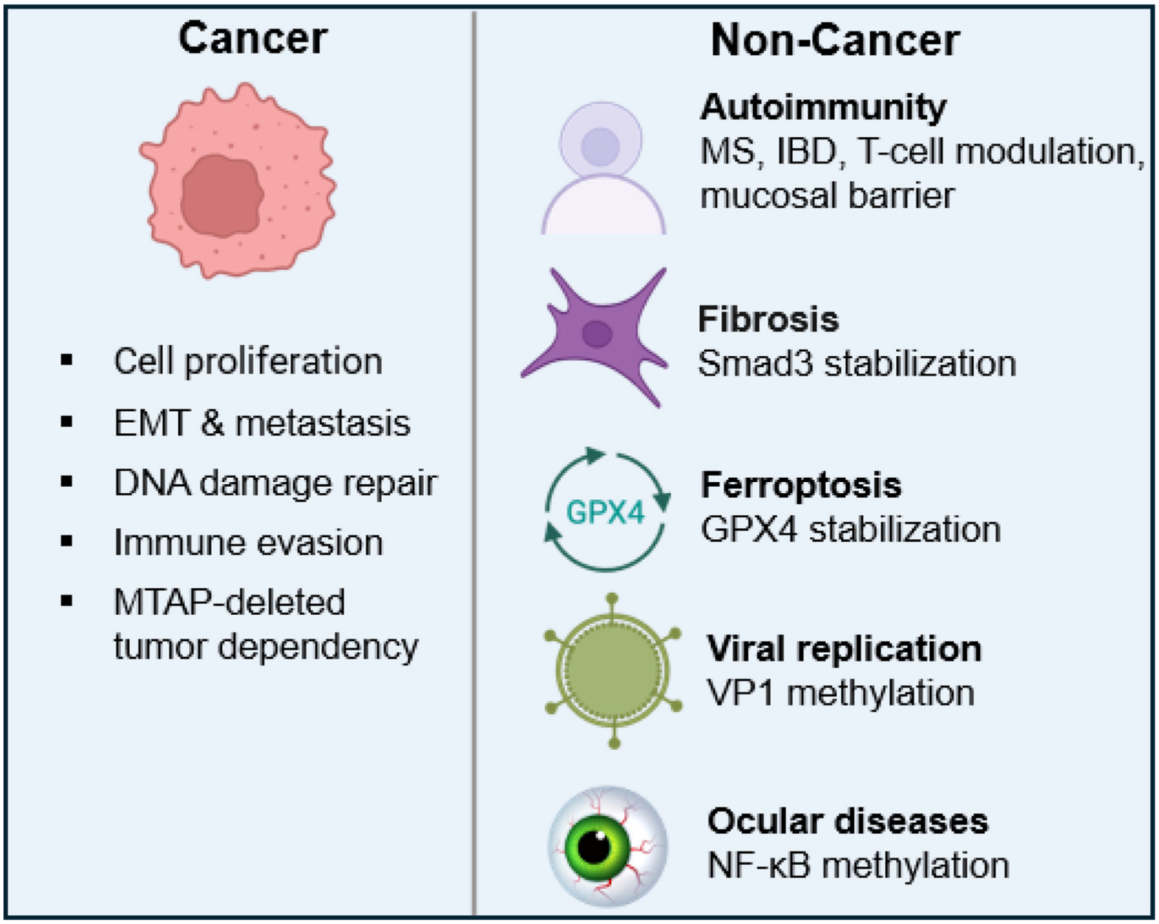

Fig. 4. Pathological Roles of PRMT5 in Cancer and Beyond.

In cancer, PRMT5 drives multiple oncogenic processes, including tumor growth, DNA damage repair, epithelial–mesenchymal transition (EMT) and metastasis, as well as immune evasion. Beyond oncology, aberrant PRMT5 activity has been implicated in diverse pathological conditions, contributing to immune dysregulation, fibrosis, resistance to ferroptosis, viral replication, and ocular disorders. Created with BioRender.com.

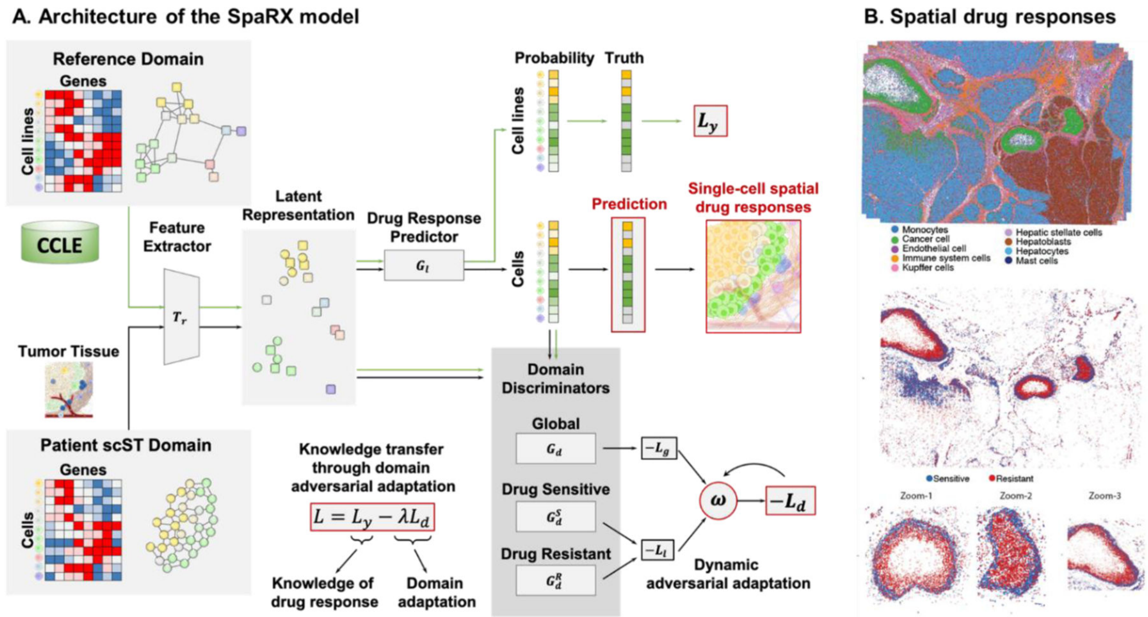

Fig. 5. SpaRx utilizes graph artificial intelligence and adversarial learning to predict spatially differential responses to drugs in liver cancer.

(A) The architecture of the SpaRX model. SpaRx transfers the knowledge from pharmacogenomics profiles to single-cell spatial transcriptomics data, through hybrid learning with dynamic adversarial adaptation.

(B) Spatial drug responses. Application of SpaRx to the state-of-the-art single-cell spatial transcriptomics data reveals that tumor cells in various locations of a tumor lesion present heterogeneous sensitivity or resistance to drugs. Moreover, resistant tumor cells interact with themselves or the surrounding constituents to form an ecosystem for drug resistance. Collectively, SpaRx characterizes the spatial therapeutic variability, unveils the molecular mechanisms underpinning drug resistance, and identifies personalized drug targets and effective drug combinations.

Table 1.

Overview of PRMT5 Inhibitor Classes, Mechanisms of Action, and Comparative Advantages/Limitations

Inhibitor Class	Mechanism of Action	Representative Inhibitors	Tumor Specificity	Clinical Stage	Advantages	Limitations	Citations
SAM-competitive inhibitors	Compete with S-adenosylmethionine (SAM) for binding to the PRMT5 cofactor site	JNJ-64619178, LLY-283, PF-06939999, PRT543, PRT811	Broad activity; effective in both MTAP+/+ and −/−	Several in clinical trials (e.g., LLY-283, JNJ64619178)	High selectivity due to unique active site conformation (e.g., Phe327, C449); potent	Risk of off-target effects on other SAM-utilizing enzymes; ATP-site selectivity challenges	[13–15]
Substrate-competitive inhibitors	Bind in the arginine substrate-binding pocket; do not interact with SAM	GSK3235025 (EPZ015666), GSK3326595 (EPZ015938)	Broad activity; effective in both MTAP+/+ and −/−	Clinical trials (e.g., GSK3326595)	Selective PRMT5 inhibition with minimal SAM interference	May face lower efficacy in MTAP-deleted cancers; resistance through substrate mimicry	[17,18]
Substrate-dependent inhibitors	Binds SAM pocket when SAM is absent and shifts to alternative site when SAM is present	PR5-LL-CM01	Broad activity; effective in both MTAP+/+ and −/−	Preclinical	Dual binding reduces off-target effects, may help circumvent resistance mechanisms affecting SAM-competitive inhibitors	Efficacy may vary depending on SAM levels and PRMT5 conformational state; Tissue-specific SAM variations could affect activity and toxicity	[8]
Dual-site (SAM + substrate) inhibitors	Engage both SAM and substrate pockets simultaneously, blocking activity from two sites	HLCL-65	Broad activity; effective in both MTAP+/+ and −/−	Preclinical	Strong inhibition; broader blockade of PRMT5	Less selective than substrate-dependent inhibitors; preclinical only	[21]
MTA-cooperative inhibitors	Exploit elevated MTA in MTAP-deleted tumors to selectively inhibit PRMT5-MEP50	AM-9747, AMG193, AZD-3470, BAY 3713372, BGB-58067, BMS-986504 (MRTX1719), CTS3497, IDE397, PEP08, S095035, TNG456, TNG462, TNG908	High specificity for MTAP-deleted cancers	Preclinical and early-stage clinical	Synthetic lethality strategy; minimal impact on normal cells	Limited applicability (~15% of cancers are MTAP−/−)	[22–27]
PROTAC degraders	Induce PRMT5 degradation via proteasome	MS4322 YZ-836P	Broad activity; effective in both MTAP+/+ and −/−	Preclinical	• Potential for deeper and longer suppression • May overcome resistance	Very early stage • PK/PD and safety not yet established	[31, 32]
Repurposed drugs	Drugs originally developed for other indications found to inhibit PRMT5	Candesartan, Cloperastine, Tadalafil	Broad activity; effective in both MTAP+/+ and −/−	Preclinical	Known safety profile; rapid repositioning possible; may sensitize tumors to chemotherapy	Typically lower potency compared to designed inhibitors; may require structural optimization	[9, 33]

Table 2.

Current Clinical Trial Landscape of PRMT5 Inhibitors (Data Source: ClinicalTrials.gov)

Company	Product	Clinical Trial	Status	Clinical indication
Amgen	AMG 193	Phase 1/1 b/2, NCT05094336 Phase 1b, NCT06333951	Recruiting	Safety, Tolerability, Maximum Tolerated Dose (MTD) in metastatic or advanced MTAP-null solid tumors. Advanced Thoracic Tumors With Homozygous MTAP-deletion
AstraZeneca	AZD3470	Phase 1/2, NCT06130553NCT06137144	Recruiting	Relapsed/Refractory Hematologic Malignancies. (PRIMAVERA), MTAP Deficient Advanced/Metastatic Solid Tumors (PRIMROSE)
Bayer	BAY 3713372	Phase 1, NCT06914128	Recruiting	Safety in MTAP-deleted Solid Tumors
BeiGene	BGB-58067	Phase 1a/b, NCT06589596	Recruiting	Safety, Tolerability, Pharmacokinetics, Pharmacodynamics, Antitumor Activity in MTA-Cooperative PRMT5 in Advanced Solid Tumor
Bristol Myers Squibb	BMS-986504 (MRTX1719)	Phase 1/2, NCT05245500, NCT06883747, NCT06672523, NCT07077434, NCT07076121, NCT07063745	Recruiting	Solid Tumors With MTAP Deletion, Recurrent GBM
CytosinLab Therapeutics	CTS3497	Phase 1/2, NCT06971523	Recruiting	MTAP Deficient Malignancies
GSK	GSK3326595	Phase 2, NCT04676516	2022-10-25 completed	Early-Stage Breast Cancer
IDEAYA Biosciences	IDE397	Phase 1, NCT04794699	Recruiting	Solid Tumors Harboring MTAP Deletion
Janssen Research & Development	JNJ-64619178	Phase 1, NCT03573310	Active, not recruiting	MTD in Advanced Solid Tumors, NHL, Lower Risk MDS
Pfizer	PF-06939999	Phase 1, NCT03854227	Terminated	Advanced Solid Tumors, Metastatic Solid Tumors
PharmaEngine	PEP08	Phase 1, NCT06973863	Not yet recruiting	MTAP-Del Advanced or Metastatic Solid Tumors
Prelude Therapeutics	PRT543, PRT811	Phase 1, NCT03886831, NCT04089449	Completed, 2023-03-28, 2023-04-05	Advanced Solid Tumors, Hematologic Malignancies, CNS Lymphoma and Gliomas
Servier Bio-Innovation	S095035	Phase 1/2, NCT06188702	Recruiting	Adult With Advanced or Metastatic Solid Tumors With Deletion of MTAP
Tango Therapeutics	TNG456, TNG462, TNG908	Phase 1/2, NCT05732831, NCT05275478, NCT06922591,	Recruiting	Safety, Tolerability in MTAP-deleted Solid Tumors